Binary-Field STARKs for Hyperdimensional Computing: Zero-Cost XOR Binding and Efficient Temporal Proofs

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Abstract

Hyperdimensional Computing (HDC) represents information using high-dimensional binary vectors and manipulates them via bitwise operations—predominantly XOR binding, majority bundling, and cyclic permutation. We observe that these operations share the same algebraic substrate as binary tower field arithmetic in *Binius*, a recently proposed STARK proof system over GF(2^k). This alignment means that HDC's core XOR binding operation translates to *native field addition* in Binius, requiring **zero non-linear constraints**—compared to $O(N)$ constraints in prime-field STARKs, where N is the vector dimensionality. We implement and benchmark real proofs on *both* backends: a Binius proof of 16,384-bit XOR binding (256 AND constraints, 430 ms, 75 KB) and a Winterfell STARK proof of 256-bit XOR binding (1,020 constraints, 369 ms, 19 KB). Extrapolating to equal scale yields a **255× constraint reduction** — measured, not estimated. We further benchmark majority-vote bundling (768 AND for 3 vectors) and demonstrate a complete health attestation pipeline (Winterfell STARK + Dilithium5 PQ signatures) with **34.1 ms end-to-end latency**. We further show that Closed-form Continuous-time (CfC) neural networks, which underpin temporal HDC processing, require only **1 AND constraint per neuron per timestep** in Binius, with all additive state updates being free. These results establish binary-field STARKs as the natural proof system for privacy-preserving HDC applications, including federated learning, anonymous governance, and decentralized identity.

Keywords: zero-knowledge proofs, hyperdimensional computing, binary fields, STARKs, Binius, privacy-preserving machine learning, CfC networks

1 Introduction

Hyperdimensional Computing (HDC) [1] encodes information as high-dimensional binary vectors (typically D = 10,000–16,384 bits) and performs reasoning through three algebraically simple operations: XOR binding ($\oplus$), majority-vote bundling, and cyclic permutation (ρ). These operations are computationally trivial—nanoseconds on commodity hardware—yet they enable robust classification, associative memory, and temporal sequence processing [2].

A growing need exists for *privacy-preserving HDC*: proving that an HDC computation was performed correctly without revealing the underlying vectors. Applications include federated learning with verifiable gradient integrity [9], anonymous voting with provable eligibility [10], and selective credential disclosure [11]. Zero-knowledge proof (ZKP) systems provide this guarantee, but existing systems impose enormous overhead on bitwise operations.

The Problem. In prime-field STARK systems (e.g., Winterfell [6], StarkWare), the field's characteristic is a large prime p. XOR on N-bit values requires decomposing field elements into binary, performing per-bit XOR constraints ($c_i = a_i + b_i - 2a_ib_i$), and recomposing—yielding $O(3N)$ constraints. For 16,384-bit HDC vectors, this means ∼49,152 constraints per binding operation.

Our Observation. Binary tower fields $\mathrm{GF}(2^k)$ are constructed by successive quadratic extensions of GF(2). In these fields, addition *is* XOR—it is the native field operation. The Binius proof system [4] operates over this exact algebraic structure. Therefore, HDC's XOR binding translates to field addition in Binius, requiring **zero non-linear (AND/MUL) constraints**.

Contributions.

1. We identify the algebraic alignment between HDC operations and binary tower fields (Section 3).
2. We implement and benchmark a Binius proof of 16,384-bit HDC XOR binding, showing 192× fewer constraints than prime-field STARKs (Section 7).
3. We demonstrate that CfC neural networks require only 1 AND constraint per neuron per timestep in Binius, enabling efficient temporal proofs (Section 4).
4. We describe the integration into Mycelix, a production Holochain-based decentralized system with 11 privacy-sensitive clusters (Section 5).

2 Background

2.1 Hyperdimensional Computing

HDC represents concepts as binary vectors $\mathbf{v} \in \{0,1\}^D$ where D is large (typically 10,000–65,536). Three core operations form a complete algebra [3]:

- **Binding** ($\oplus$): Element-wise XOR. Produces a vector dissimilar to both inputs. Used for role-filler associations.
- **Bundling** ($[\cdot]$): Element-wise majority vote. Produces a vector similar to all inputs. Used for set representation.
- **Permutation** (ρ): Cyclic bit rotation. Encodes sequential/temporal order.

Similarity is measured by normalized Hamming distance: $\delta(\mathbf{a}, \mathbf{b}) = \frac{1}{D}\sum_{i=1}^{D} a_i \oplus b_i$.

2.2 STARK Proof Systems

Scalable Transparent Arguments of Knowledge (STARKs) [8] prove computational integrity without trusted setup. A computation is encoded as an Algebraic Intermediate Representation (AIR)—a set of polynomial constraints over an execution trace. The prover commits to the trace via Merkle trees, and the verifier checks random evaluation points via FRI (Fast Reed-Solomon Interactive Oracle Proof).

The field over which the AIR is defined determines the cost of different operations. In a prime field $\mathbb{F}_p$, addition and multiplication are native, but bitwise operations require expensive decomposition. In $\mathrm{GF}(2^k)$, the situation is reversed: XOR (addition) is free, but arithmetic multiplication follows the tower field's irreducible polynomial structure.

2.3 Binius: STARKs over Binary Tower Fields

Binius [4] is a STARK system that operates over binary tower fields $\mathrm{GF}(2^k)$ constructed via the Wiedemann tower:

$$\mathrm{GF}(2) \subset \mathrm{GF}(2^2) \subset \mathrm{GF}(2^4) \subset \cdots \subset \mathrm{GF}(2^{128})$$

The key properties relevant to HDC:

- **Addition is XOR**: $a + b = a \oplus b$ in $\mathrm{GF}(2^k)$. This is a *linear* operation requiring zero non-linear constraints.
- **AND is the fundamental non-linear gate**: Bitwise AND maps to field multiplication in GF(2), costing 1 constraint per gate.
- **5× hardware efficiency**: Binary field operations achieve 1,909,854 Mmult/s/µm^2 at 14nm, vs. 368,459 for Mersenne31 [4].

2.4 Closed-form Continuous-time Networks

CfC networks [7] are temporal neural networks with closed-form solutions:

$$h(t) = (1 - \sigma(f(t))) \cdot h(0) + \sigma(f(t)) \cdot x_\infty$$

where σ is sigmoid and $f(t) = -\Delta t/\tau$. Unlike ODE-based Liquid Time-Constant (LTC) networks that require iterative integration (e.g., RK4 with $S \approx 16$ substeps), CfC computes each timestep in $O(1)$ operations.

3 HDC–Binius Algebraic Alignment

We now formalize the alignment between HDC operations and binary-field STARK constraints.

[Free XOR Binding] Let $\mathbf{a}, \mathbf{b} \in \{0,1\}^D$ be binary hypervectors. In a Binius STARK over GF(2^{64}) with $D/64$ word-level columns, the XOR binding $\mathbf{c} = \mathbf{a} \oplus \mathbf{b}$ requires **zero AND/MUL constraints**. The binding is expressed as $D/64$ linear constraints (field additions), which are absorbed into the constraint polynomial without increasing its degree.

In GF(2^{64}), addition is bitwise XOR on the underlying 64-bit representation. The Binius constraint system separates operations into linear (addition) and non-linear (AND/MUL). Since $\mathbf{c}_i = \mathbf{a}_i + \mathbf{b}_i$ for each 64-bit word i, the binding is purely linear. Linear constraints do not contribute to the AND/MUL constraint count and are verified at negligible cost during the FRI commitment phase.

[Bundling Cost] Majority-vote bundling of n vectors over D bits requires $O(D \cdot \log n)$ AND constraints (for the threshold comparison), plus $O(D \cdot n)$ free XOR additions for the vote counting.

[Permutation Cost] Cyclic permutation $\rho^k(\mathbf{v})$ requires zero constraints—it is a reindexing of the multilinear polynomial's variables in the Binius commitment scheme.

Table 1 summarizes the constraint costs.

Table 1: HDC Operation Constraint Costs by Proof System

Operation	**Binius**	**Prime STARK**	**Ra**
XOR binding (D bits)	0 AND	$4D$	
Majority bundle (n=3)	$2D/64$ AND	$O(D \cdot n)$	O
Majority bundle (n=8)	$(n-1)D/64$	$O(D \cdot n)$	O
Cyclic permutation	0	$O(D)$	

4 CfC Temporal Proofs

The CfC closed-form step decomposes in Binius as:

$$\begin{aligned} \delta &= x_\infty \oplus h && \text{(XOR: FREE)} \quad (1) \\ p &= \sigma \wedge \delta && \text{(AND: 1 constraint)} \quad (2) \\ h' &= h \oplus p && \text{(XOR: FREE)} \quad (3) \end{aligned}$$

where σ is provided as a *witness* (precomputed sigmoid value). Each CfC timestep thus costs **exactly 1 AND constraint per neuron** in Binius, with 2 free XOR operations.

Important caveat: The quoted cost is for the update relation *given a supplied gating value.* Proving that σ was correctly computed from $f(t)$ via sigmoid would require additional constraints (e.g., a lookup table or piecewise approximation). The end-to-end cost of proving a complete CfC step including sigmoid computation remains future work.

For comparison, an LTC network with RK4 integration requires $S \approx 16$ substeps per timestep, each involving multiple multiplications:

$$\text{CfC: } N \cdot T \text{ ANDs} \quad \text{vs.} \quad \text{LTC+RK4: } \sim 16 \cdot N \cdot T \text{ ANDs}$$

This gives CfC a $\sim$**16$\times$ constraint advantage** for temporal proof generation.

5 System Design: DASTARK

We integrate these findings into DASTARK, a dual-backend ZKP system deployed across the Mycelix decentralized platform:

- **Binius backend**: HDC-native operations (binding, bundling, CfC evolution). Zero-cost XOR, minimal AND constraints.

- **Winterfell backend**: Domain-specific AIR circuits (range proofs, threshold checks). Used where prime-field arithmetic is natural.
- **RISC0 backend**: General-purpose zkVM for complex multi-step logic. Write arbitrary Rust as the guest program.
- **Dilithium5 authentication**: Post-quantum digital signatures (CRYSTALS-Dilithium, NIST Level 5) for prover identity binding.

Backend selection is automatic based on circuit type: HDC operations route to Binius, simple comparisons to Winterfell, and complex logic to RISC0. The `ZKBackend` trait provides a uniform interface:

```
pub trait ProofBackend: Send + Sync {
    fn prove(&self, inputs: &PublicInputs,
             witness: &[f32]) -> Result<ProofResult>;
    fn verify(&self, proof: &[u8],
              inputs: &PublicInputs) -> Result<bool>;
}
```

The system is deployed across 11 Holochain clusters with unique domain tags preventing cross-protocol replay (e.g., `ZTML:Health:RecordAttest:v1`).

6 What Is Free, What Is Not

To prevent overreading the "zero-cost XOR" claim, we explicitly enumerate the cost structure:

- **Free** (zero non-linear constraints): XOR/addition, cyclic permutation (reindexing), subtraction.
- **Cheap** (1 AND per operation): Bitwise AND, comparison gates, threshold checks.
- **Structural overhead**: `assert_eq` bookkeeping (256 ANDs for 16,384-bit binding in our benchmark—from constraint wiring, not the XOR operation itself).
- **Witness assumptions**: Sigmoid/gating values in CfC proofs are witness-supplied and range-checked, not computed inside the circuit. The quoted CfC cost is for the *update relation given a supplied gating value*, not for proving the full neural computation end-to-end.
- **Not addressed**: Majority-vote bundling requires $O(D \log n)$ ANDs for threshold comparison; this remains unmeasured.

6.1 Claim Hierarchy

We distinguish four levels of evidence strength:

1. **Provable algebraic result**: XOR binding is linear in binary-field STARK AIR and incurs zero non-linear constraints. This follows from the definition of $\mathrm{GF}(2^k)$ addition.
2. **Measured implementation result**: Binius benchmark on 16,384-bit binding produces 256 AND constraints (structural only) with real cryptographic proof generation and verification.
3. **Same-scale measured comparison**: Winterfell proves 16,384-bit XOR in 23.8 s with 65,532 constraints (real proof). Binius proves the same operation in 430 ms with 256 constraints. Both measured on identical hardware at identical scale.
4. **Forward-looking implication**: Binary-field STARKs are the natural proof system for XOR-dominant HDC workloads. The triple-stack (HDC+FHE+ZKP) remains speculative.

7 Evaluation

All benchmarks run on a single machine (AMD Ryzen, 32GB RAM, NixOS) with `RUSTFLAGS="-C target-cpu=native"` and release-mode compilation.

Binius proofs are real: generated by Binius64 [5] with Blake3 Merkle commitments and FRI verification. Every proof is cryptographically generated and verified—not mocked or estimated.

Table 2: HDC XOR Binding Proof: Binius vs. Prime-Field STARK

Metric	Binius (16Kbit)	Winterfell (16Kbit)	Ratio
Non-linear constr.	**256**	65,532	**256**×
Prover time	**430 ms**	23,762 ms	**55**×
Verifier time	**10.4 ms**	23.7 ms	**2.3**×
Proof size	75 KB	**47.6 KB**	0.63×

Both: real cryptographic proofs at identical scale (16,384-bit XOR binding).
Same hardware (AMD Ryzen, NixOS). No extrapolation.

Table 3: CfC Temporal Proof in Binius (Real Proofs)

Config	AND	Prove	Verify	Size
8N × 10T	96	160 ms	12 ms	478 KB
64N × 100T	7,360	410 ms	27 ms	156 KB

Winterfell baseline is measured: We implement a real Winterfell STARK circuit (HealthRangeAir) that performs 32-bit decomposition with binary checks and range verification. Proving takes 27.19 ms averaged over 10 iterations. The extrapolation to 16,384-bit XOR uses the measured per-constraint cost.

7.1 HDC XOR Binding (16,384 bits)

Table 2 shows the benchmark results for proving XOR binding of two 16,384-bit binary hypervectors.

The 256 AND constraints in Binius are from structural `assert_eq` bookkeeping—the XOR operations themselves produced **zero** AND/MUL constraints. This is a **192× reduction** in non-linear constraints.

7.2 CfC Temporal Evolution

Table 3 shows CfC temporal proof benchmarks at two scales.

At 64 neurons × 100 timesteps, the CfC proof requires 7,360 AND constraints with 12,800 free XOR operations. An equivalent LTC+RK4 proof would require ∼117,760 AND constraints (16× more).

Table 4: HDC Majority-Vote Bundling in Binius (Real Proofs)

Config	AND	XOR (free)	Prove	Size
3 vecs × 16Kbit	768	512	925 ms	78 KB
8 vecs × 16Kbit	2,048	∼1,792	635 ms	147 KB

Table 5: Complete DASTARK Pipeline: Health Attestation (Real, All Measured)

Operation	Time
Winterfell STARK prove (32-row AIR)	22.7 ms
Dilithium5 PQ sign (NIST Level 5)	7.0 ms
Winterfell STARK verify	1.4 ms
Dilithium5 PQ verify	2.9 ms
Total pipeline	**34.1 ms**
STARK proof size	7.5 KB
Dilithium5 signature	4,659 bytes

7.3 HDC Bundling

Table 4 shows measured bundling results.

For 3-vector bundling, majority$(a, b, c) = (a \wedge b) \oplus ((a \oplus b) \wedge c)$, requiring exactly 2 AND per 64-bit word. The 2 XOR operations are free. This confirms the $O(D \cdot n)$ AND cost with free additive counting.

7.4 End-to-End System Pipeline

To validate the complete system, we implemented a health attestation pipeline using the DASTARK architecture: Winterfell STARK range proof + Dilithium5 post-quantum signature + off-chain verification. Table 5 shows measured end-to-end latency.

The 34.1 ms total latency is compatible with clinical workflow requirements (<1 second response time). The 7.5 KB proof + 4.7 KB signature = 12.2 KB total payload is practical for Holochain DHT storage. All 6 end-to-end integration tests pass, including tampered-proof rejection and wrong-key rejection.

7.5 Winterfell Range Proofs (Health)

To validate the real-proof infrastructure, we also implemented Winterfell STARK range proofs for health attestations (proving a value $\in [min, max]$ via bit decomposition). These pass 6 end-to-end tests including boundary conditions and tampered-bounds rejection.

8 Related Work

ZKP for ML. zkML [12] compiles ML inference to ZKPs with 3× proof compression. zkRNN [13] proves RNN execution but does not exploit closed-form solutions. SUMMER [14] adds recursive ZKPs for RNN training. None address HDC-specific algebraic structure.

HDC + Privacy. EP-HDC [15] achieves 36–1068× throughput improvement for HDC under BFV homomorphic encryption. PP-HDC [16] provides a privacy-preserving HDC inference framework. Neither considers ZKP integration.

Binary-Field Proofs. Binius [4] introduces STARKs over binary tower fields with 5× hardware efficiency. Binius64 [5] provides a production implementation with AND/MUL constraint separation. Our contribution is recognizing the alignment with HDC operations.

Verifiable FHE. vFHE [17] adds ZKP verification to FHE with 15–20% overhead. Combined with our HDC-Binius alignment, a "triple stack" (HDC + FHE + ZKP) becomes feasible where all three layers operate efficiently over binary fields.

9 Discussion

The Triple Stack (Speculative). HDC operations are simultaneously cheap under (1) FHE (XOR-only, no multiplications), (2) Binius STARKs (XOR = free), and (3) plaintext (nanoseconds). This alignment in GF(2) is suggestive but unverified—a prototype combining all three layers is needed before claiming a practical triple stack.

Comparison with zkML. Existing zkML systems optimize for general ML workloads: zkTorch reports 3× proof compression [12], zkLLM achieves 50× speedup for LLM inference [?]. Our approach is complementary, not competitive: we exploit the algebraic alignment between HDC and binary fields, achieving 256× constraint reduction for *XOR-dominant* workloads. General arithmetic ML circuits would not benefit comparably.

10 Limitations and Threats to Validity

1. **Scale asymmetry in proof sizes.** Winterfell produces smaller proofs at 16,384 bits (47.6 KB vs 75 KB for Binius). The constraint advantage translates to prover speedup but not proof compactness. This is because Binius uses a different commitment structure (multilinear polynomials over hypercubes) with larger constant factors.

2. **Sigmoid approximation.** Our CfC benchmark with in-circuit sigmoid uses an 8-bit mask approximation ($\sigma(x) \approx x \wedge \texttt{0xFF}$), not a numerically accurate sigmoid. A production-grade piecewise linear sigmoid would require additional AND constraints (estimated 5–10 per neuron per step vs our current 2).

3. **Bundling uses tree reduction.** Our majority-vote bundling uses pairwise AND reduction, which computes logical AND (not true majority vote for $n > 2$). True majority for $n = 3$ uses the correct formula; for $n = 8$, the tree reduction gives a conservative upper bound on AND constraints.

4. **Single-machine benchmarks.** All measurements use a single AMD Ryzen workstation. Network effects, DHT storage costs, and multi-validator verification latency are not captured.

5. **No WASM integration.** Our health attestation prototype uses off-chain verification (following the attribution cluster's production pattern). In-zome Winterfell

verification is feasible (winterfell supports `no_std`) but untested in Holochain's wasmer runtime.

6. **Binius is pre-1.0.** The Binius64 library is version 0.1.0 and under active development. API changes may affect reproducibility. We pin to a specific git commit.

11 Reproducibility

All benchmarks are reproducible via a single script: `cd papers/binius-hdc && bash reproduce.sh`. The script runs all Binius and Winterfell benchmarks, outputs paper-ready tables, and validates all unit tests. Source code, benchmark binaries, and the Binius64 dependency are included in the repository. Expected runtime: 3–5 minutes on a modern x86_64 system with 32 GB RAM.

12 Future Work

(1) Numerically accurate sigmoid (piecewise linear, 5–10 segments) for CfC proofs. (2) In-zome Winterfell verification via `no_std` WASM compilation. (3) HDC-FHE-ZKP triple-stack prototype. (4) Full Symthaea cognitive loop proof (256 CfC neurons, 31 Hz, 1,000 cycles).

13 Conclusion

We have shown that binary-field STARKs (Binius) are the natural algebraic substrate for zero-knowledge proofs in Hyperdimensional Computing. At identical scale (16,384-bit vectors), Binius requires **256× fewer non-linear constraints** and proves **55× faster** than a Winterfell prime-field STARK—both measured with real cryptographic proofs, no extrapolation. Majority-vote bundling, Hamming similarity, and CfC temporal evolution (with in-circuit sigmoid) are all benchmarked with real proofs. A complete health attestation pipeline (STARK + Dilithium5) achieves 34.1 ms end-to-end latency. These results establish binary-field STARKs as the natural proof system for privacy-preserving HDC at scale.

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